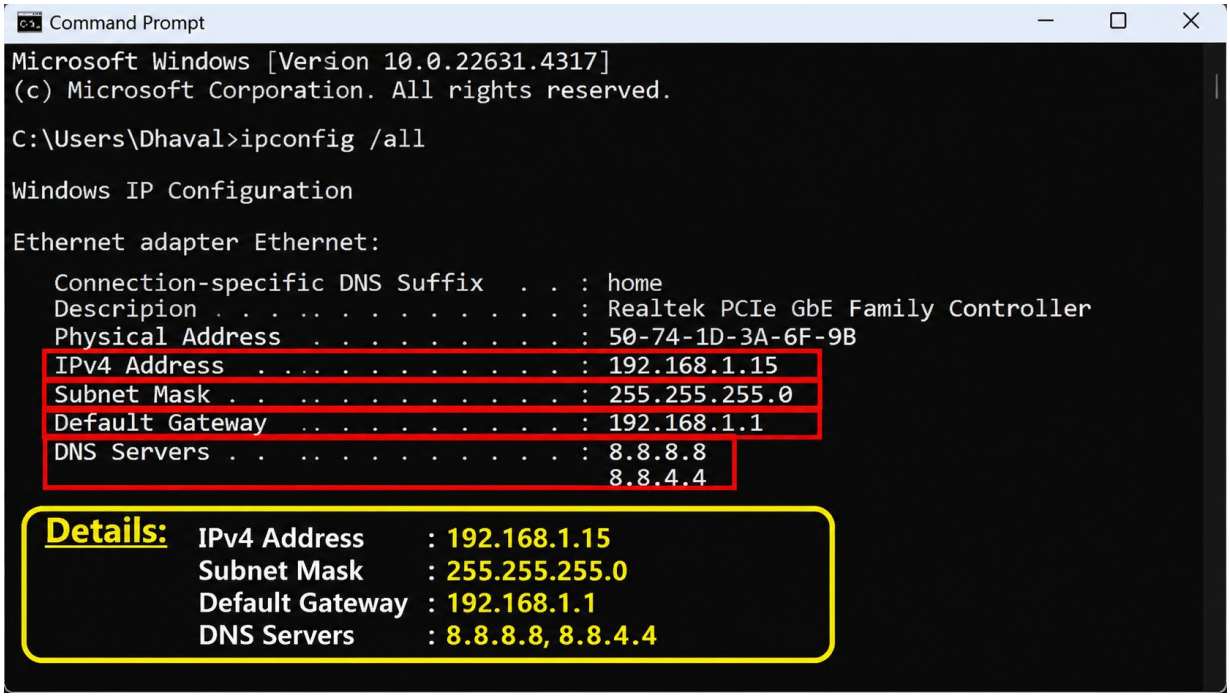
Assignment – 2.8

Network Troubleshooting, Management, Monitoring, and Optimization

Q - 1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.



Q - 2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

- **Check Network Connectivity Using Ping Command:**
- **Difference Between Results:**

Without Internet	With Internet
Ping fails	Ping succeeds
No reply received	Reply received
Network is disconnected	Network is connected

- **Meaning of Each Status:**

1. **Reply from:** The destination is reachable.
2. **Request timed out:** No reply was received within the time limit.
3. **Destination host unreachable:** The network cannot reach the destination.
4. **Could not find host:** The domain name could not be resolved.

Q - 3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

➢ Monitor Network Usage:

- **Peak Network Usage:** Depends on internet speed and video quality.
- **Process Used Most Bandwidth:** Google Chrome (if YouTube is opened in Chrome).

Q - 4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets. Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.

- **Wireshark Packet Capture:**
- **Three types of information visible in packets:**

1. **Source IP Address:** Shows the IP address of the sending device.
2. **Destination IP Address:** Shows the IP address of the receiving server.
3. **Protocol Information:** Shows whether the communication uses HTTP or HTTPS (TLS).

Q - 5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

- **Troubleshooting Checklist for Slow Internet:**
- **AI-Generated Response:**

1. Restart the Wi-Fi router.
2. Use an Ethernet cable.
3. Check internet speed.
4. Close unnecessary background apps.
5. Keep the device near the router.
6. Disconnect unused devices from Wi-Fi.
7. Update the router software.
8. Contact the internet provider if the problem continues.

- **My First Choice:**

I would first try **restarting the Wi-Fi router** because it may improve internet speed and connection.